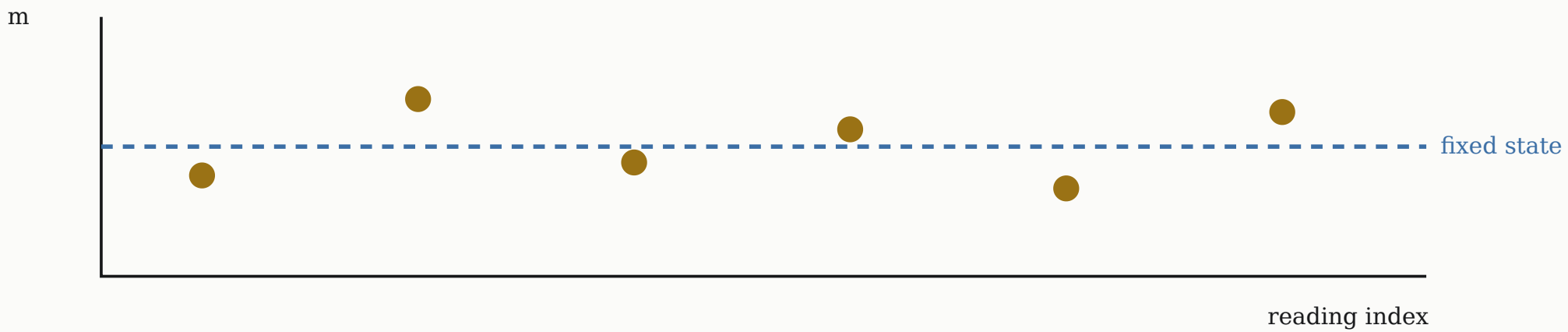
MOTIVATION

From Noisy Readings to Reliable Estimates

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

Report an estimate and the uncertainty around it.

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon · Jing Zhang · Fall 2026 · November 4

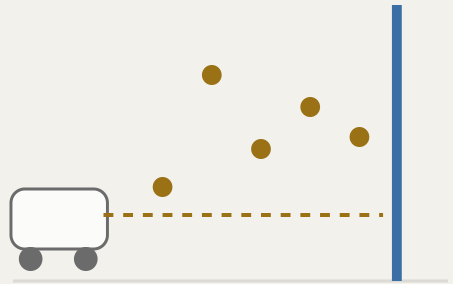
Probability model; parameter estimation — [1] DL §3.1 · [1] DL §5.5.1 · Course example

ROADMAP

From Readings to Estimates

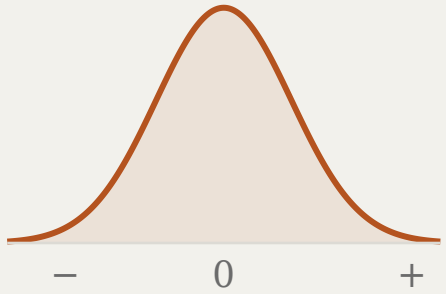
How does the lecture fit together?

Readings



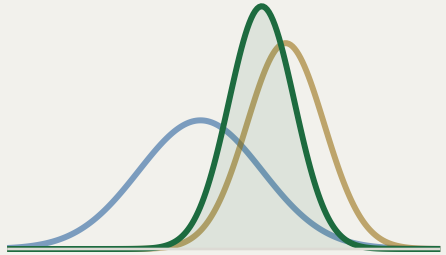
Same wall, noisy readouts

Error



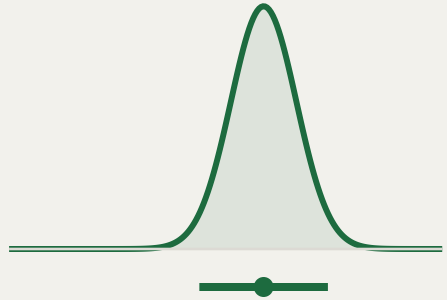
Model the error

Update



Combine information

Estimate



Report a value and a range

> What should we report? How sure are we?

Probability distribution; uncertainty — [1] DL §3.1 · Course concept map

DEFINITION

Random variables

A random variable assigns a real number to each outcome in the sample space.

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

- Ω sample space: all possible outcomes
- ω one outcome
- x random error (cm)

Outcome A $\to -1$ cm

Outcome B $\to 0$ cm

Outcome C $\to 2$ cm

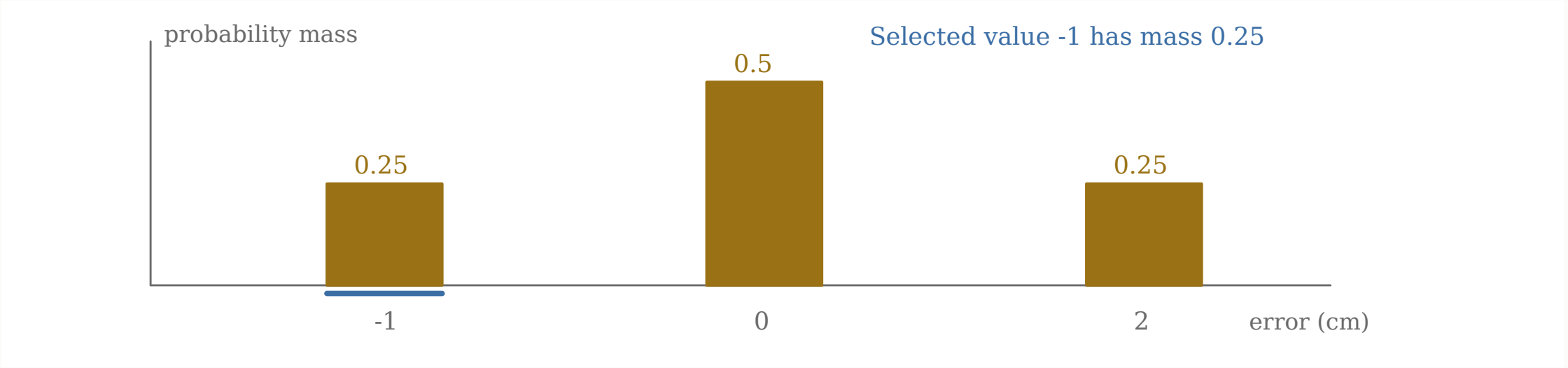
> A constant random variable assigns every outcome the same number.

Random variable; realization — [10] MC Probability §5.2

DEFINITION

Probability Mass Function (PMF)

A PMF assigns a probability to each value of a discrete random variable.



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

- x discrete error: finitely or countably many values (cm)
- a possible value (cm)
- $p(a)$ probability of $x = a$
- P event probability; an event is a set of outcomes

Probability mass function (PMF) — [1] DL §3.3.1 · [7] Data140 §1.1

EXAMPLE

Probability Mass Function (PMF)

The mean is the probability-weighted average error.

Error values −1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = \sum_a a\, p(a) = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- $a, p(a)$ possible error (cm), its probability mass
- P event probability
- x error (cm)
- μ mean error; sensor bias (cm)

Discrete probability; expectation — [1] DL §3.3.1 · [7] Data140 §§8.1,8.3 · Course example

EXERCISE

Probability Mass Function (PMF)

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

DEFINITION

Probability Density Function (PDF)

A PDF gives interval probabilities as areas under a nonnegative density curve.

$$p(t) \geq 0, \quad \int_{-\infty}^{\infty} p(t)\,dt = 1$$

$$P(a \leq x \leq b) = \int_{a}^{b} p(t)\,dt$$

- x random error (cm)
- a, b interval endpoints, $a \leq b$
- t integration variable
- $p(t)$ density (cm^{-1})

$$P(x = c) = \int_{c}^{c} p(t)\,dt = 0$$

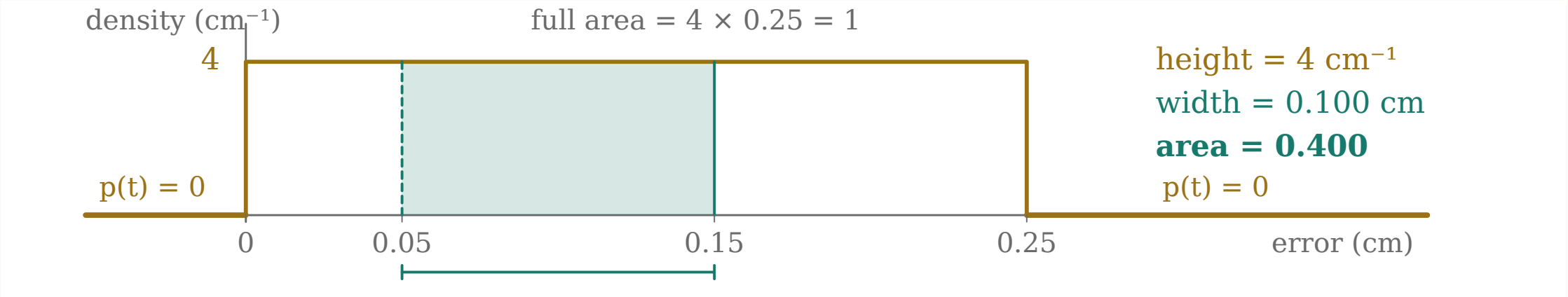
- c fixed error value

Probability density function (PDF) — [1] DL §3.3.2

EXAMPLE

Probability Density Function (PDF)

A uniform density is constant on its interval and zero outside it.



UNIFORM ERROR

$$p(t) = \begin{cases} 4 & 0 \le t \le .25 \\ 0 & \text{otherwise} \end{cases}, \qquad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

x random error (cm) t density argument (cm) p density (cm^{-1})

P interval probability

Uniform density; normalization — [1] DL §3.3.2 · Course example

EXERCISE

Probability Density Function (PDF)

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is $.2$, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

DEFINITION

Moments

Expectation is the probability-weighted average of a function’s values.

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete variable
- a possible value
- g function; assume $\mathbb{E}[|g(x)|] < \infty$
- p PMF; moments average powers: mean $\mathbb{E}[x]$, second moment $\mathbb{E}[x^2]$

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous variable
- t integration variable
- p density
- g same function

Expectation of a function — [1] DL §3.8 · [7] Data140 §§8.1,8.3 · Course derivation

EXERCISE

Moments

Variance is the average squared deviation from the mean.

$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$. Given $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

DEFINITION

Moments

The mean vector lists the component expectations.

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x 2×1 random column on one outcome space
- x_i scalar coordinate selected by i
- μ fixed 2×1 mean; finite component means

> Average components; keep the column shape.

Sample index s: $x^{(s)}$. Component index i: x_i.

Random vector; mean vector — [7] Data140 §23.1

DEFINITION

Moments

Covariance averages products of deviations from the component means.

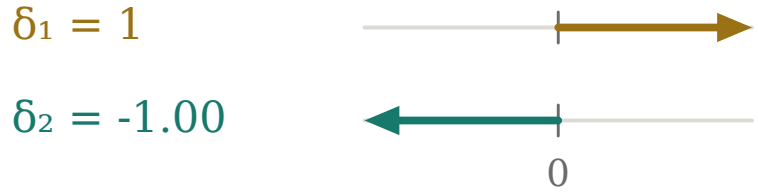
$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

x, μ d×1 random column / fixed mean; d: dimension; $\mathbb{E}[x_i^2] < \infty$

Σ population covariance, d × d i, j coordinate indices

One centered outcome: δ = x − μ

dimensionless illustration



One outer product $\delta\delta^T$

$\delta_1^2 = 1$	$\delta_1\delta_2 = -1.00$
$\delta_2\delta_1 = -1.00$	$\delta_2^2 = 1.00$

Σ averages these matrices over the population.

> In Σ: diagonal = component variance; off-diagonal = mean paired product.

Covariance matrix — [1] DL §3.8 · [7] Data140 §23.1 · Course derivation

DERIVATION

Moments

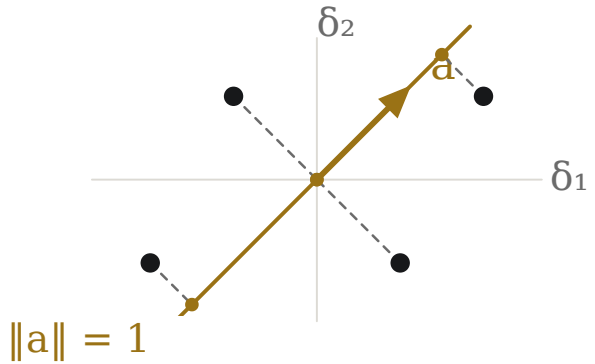
Positive semidefinite means symmetric, with $a^T \Sigma a \geq 0$ for every a.

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed d×1 direction; d = dimension
- x, μ d×1 random column / fixed population mean
- Σ d×d population covariance (PSD)

$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C d×d sample covariance estimate
- X_c N×d matrix: N centered rows, d coordinates
- N sample count; N>1



Centered population: each point has probability ¼

δ = x − μ	y = aᵀδ	y² ≥ 0
(2, 1)	2.12	4.50
(-2, -1)	-2.12	4.50
(1, -1)	0.00	0.00
(-1, 1)	0.00	0.00

Var(y) = average of y² = 2.250 ≥ 0

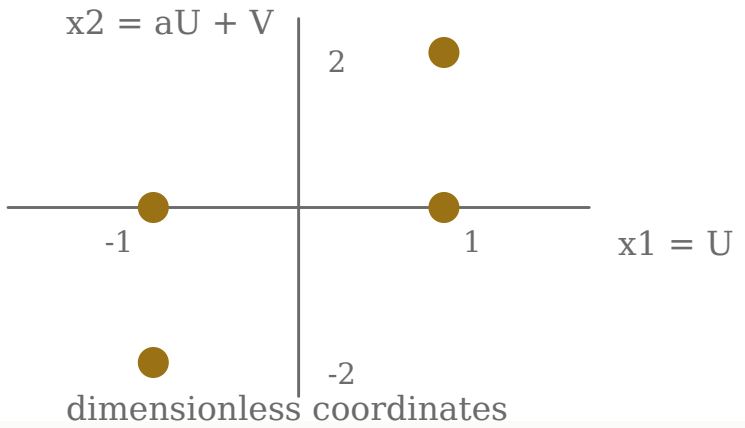
dimensionless model

Positive semidefinite covariance — [1] DL §3.8 · [1] DL §2.7 · [7] Data140 §23.1 · Course derivation

EXAMPLE

Moments

Independence means knowing one sign does not change the other's probabilities.



a = 1.00

four equally likely outcomes

a: scalar coefficient, 0 to 1

Independent fair signs: $P(U=u,V=v)=P(U=u)P(V=v)$ for all $u,v=\pm 1$. Let $x=(U,U+V)^T$.

$$\mu=\begin{bmatrix}0\\0\end{bmatrix},\qquad \Sigma=\begin{bmatrix}1&1\\1&2\end{bmatrix}$$

U,V fair signs: ±1, each with mass ½

x,μ 2×1 random column / fixed mean

Σ 2×2 covariance; a nonzero off-diagonal entry means correlation

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course example

EXERCISE

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

A. 2: both components share U.

B. 0: independent sources imply independent outputs.

C. 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

COUNTEREXAMPLE

Moments

Uncorrelated means zero covariance; independence is a stronger condition.

x is uniform on $\{-1,0,1\}$; $y=x^2$. Is y independent of x?

$$\mathrm{Cov}(x,y)=0, \qquad P(y=0 \mid x=0)=1 \neq \tfrac{1}{3}=P(y=0)$$

x, y	random scalars
$P(y=0 \mid x=0)$	probability given $x=0$: $P(y=0,x=0)/P(x=0)$
Cov	population covariance

Uncorrelatedness; independence — [1] DL §3.8 · [2] UBC Lec.18 p.8 · Course example

DEFINITION

Gaussian geometry

A vector is Gaussian if every linear combination is normal (constant at zero variance).

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-d_M^2/2)$$

- x, μ — d×1 evaluation point / fixed mean
- d — dimension
- $\Sigma \succ 0$ — d×d covariance; $a^T\Sigma a > 0$ for every nonzero direction column a
- $|\Sigma|$ — determinant
- p — density
- d_M^2 — squared Mahalanobis distance (dimensionless)

$$d_M^2 = (x-\mu)^T\Sigma^{-1}(x-\mu)$$

- Σ^{-1} — d×d inverse covariance
- $x - \mu$ — d×1 displacement from the mean

Multivariate Gaussian distribution — [1] DL §3.9.3 · [7] Data140 §23.2

DERIVATION

Gaussian geometry

Whitening centers a vector and gives it identity covariance.

Derivation

Derive the mean and covariance

x, z, μ: $d \times 1$ columns; d: dimension. Σ: covariance; Q: orthogonal eigenvector matrix; Λ: positive diagonal eigenvalue matrix. Fixed $d \times d$ whitening map $W = \Lambda^{-1/2} Q^T$; $z = W(x - \mu)$. $\mathbb{E}$: expectation; Cov: covariance; I : $d \times d$ identity.

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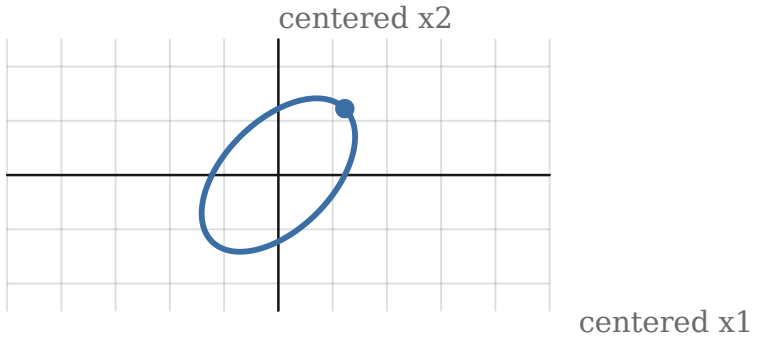
1. $\mathbb{E}[z] = W(\mathbb{E}[x] - \mu) = 0$ by linearity.
2. $\mathrm{Cov}(z) = \mathbb{E}[zz^T] = W\,\mathbb{E}[(x-\mu)(x-\mu)^T]\,W^T = W\Sigma W^T$.
3. $= \Lambda^{-1/2} Q^T (Q \Lambda Q^T) Q \Lambda^{-1/2} = \Lambda^{-1/2} \Lambda \Lambda^{-1/2} = I$.
4. Fixed matrices pass through expectation; $Q^T Q = I$ cancels the basis changes.

Whitening; eigendecomposition — [6] Stanford EE278 pp.3–10–12 · [2] UBC Lec.18 p.5 · Course derivation

DERIVATION

Gaussian geometry

A density contour joins points with the same density.



unit Mahalanobis contour

eigenvalues 3 and 1

$$x = \mu + Q\Lambda^{1/2} z, \qquad \|z\| = c$$

- x, μ — 2×1 point and fixed mean (original coordinate units).
- $Q = [q_1\ q_2]$ — fixed 2×2 orthogonal matrix; orthonormal eigenvector columns. $Q^T Q = I_2$; I_2: 2×2 identity.
- Σ, Λ — fixed covariance $\Sigma = Q\Lambda Q^T$; $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$ contains positive eigenvalue variances (squared units).
- z, c, q_i — z: dimensionless 2×1 coordinate point; $c = \|z\| > 0$: fixed dimensionless radius. q_i: unit direction; $c\sqrt{\lambda_i}$: semiaxis length in original units; $i = 1, 2$.

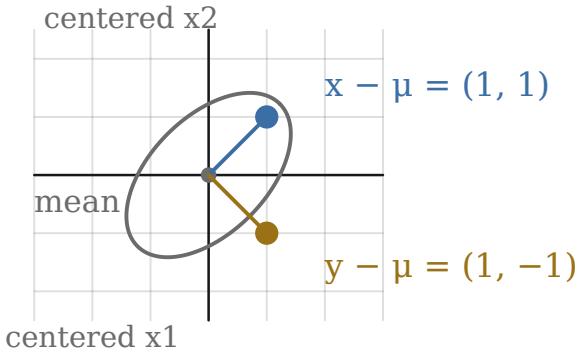
Plot of $x - \mu$ — $c = 1$, $\Lambda = \mathrm{diag}(3, 1)$; $q_1 = (1, 1)^T/\sqrt{2}$, $q_2 = (1, -1)^T/\sqrt{2}$.

Gaussian density contour — [2] UBC Lec.18 p.5 · Course derivation

EXAMPLE

Gaussian geometry

Equal Euclidean lengths can have different Gaussian densities.



gray ellipse: $d_m = 1$

both arrows: Euclidean length $\sqrt{2}$

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

x, y, μ fixed 2×1 evaluation points and mean (coordinate units); $x = (2, 0)^T$, $y = (2, -2)^T$.

Σ, d_M fixed 2×2 covariance (squared units); d_M: dimensionless covariance-scaled distance from μ. Plot axes show centered coordinates.

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

d_M covariance-scaled distance.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course example

EXERCISE

Gaussian geometry

Which point has the higher Gaussian density?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

u, v: fixed 2×1 points; μ: fixed mean column; Σ: fixed 2×2 covariance; p: density (inverse squared coordinate units). Coordinates use original units; covariance entries use squared units.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

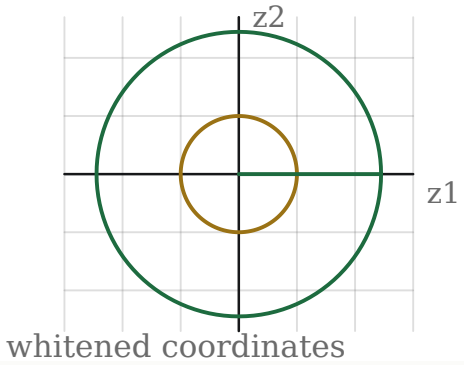
C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

DERIVATION

Gaussian geometry

Choose a region by the probability it encloses.



c = 1: mass 39.3%

c = 2.448: mass 95%

original semiaxes: 4.24 and 2.45

circle radius = Mahalanobis distance

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

z, d_M, c, d $z = (z_1, z_2)^T$: dimensionless whitened random column; $d_M = \|z\|$: dimensionless distance; $c \ge 0$: fixed dimensionless radius; $d = 2$: dimension.

P, λ_1, λ_2 P: event probability (no units). Eigenvalues $\lambda_1 = 3, \lambda_2 = 1$ are variances in original squared units. The region is $d_M \le c$; its probability is the density integral inside.

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

$c_{.95}$ radius enclosing 95%.

Chi-square distribution; Gaussian probability region — [3] NIST §1.3.6.6.6 · [3] NIST χ^2 table · Course derivation

DEFINITION

Bayes

Conditional probability is the chance of an event given another.

joint events	positive +	negative −	row total
event H	H and +	H and −	P(H)
no event	not H and +	not H and −	P(not H)
	total P(+)	total P(−)	

$$P(H \mid +) = \frac{P(H \cap +)}{P(+)} \qquad P(+ \mid H) = \frac{P(H \cap +)}{P(H)}$$

$H, +, -, P$ H: event present; $+/-$: positive/negative reading; P: event probability (no units). $|$: given; $\cap$: both occur.

Two conditions Sensitivity $P(+ \mid H)$: positive rate among event cases. Posterior $P(H \mid +)$: event probability after observing a positive. Denominators must be positive.

Conditional probability — [1] DL §3.4–3.5 · [9] Wisconsin Math225 Lec.5 · Course example

DERIVATION

Bayes

Bayes’ rule updates a prior probability using an observation.

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

$H, +, P$ H: event of interest; $+$: observed positive reading; P: event probability (no units); $|$: given.

Four roles $P(H)$ prior: chance before this reading; $P(+ \mid H)$ likelihood: positive-reading chance if H holds; $P(+) > 0$ evidence: overall positive-reading chance; $P(H \mid +)$ posterior: event chance after this reading.

Bayes’ rule — [1] DL §3.11 · [8] CMU Bayes notes p.1 · Course derivation

DERIVATION

Bayes

Total probability adds contributions from disjoint cases.

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

$H, \neg H, +, P$ event, no event, positive reading and probability (no units); $\neg$: not; $|$: given.

Partition H and $\neg H$ are disjoint and cover all trials. Their joint events with $+$ cover every positive reading. $P(+)$ is the evidence, the total positive-reading probability.

Law of total probability; evidence — [1] DL §3.4–3.5 · [1] DL §3.11 · Course derivation

EXAMPLE

Bayes

Why can most positive alarms be false?

joint events	positive +	negative −
event H	90	10
no event	990	8910

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

$H, \neg H, +, P$ event present, event absent, positive reading and probability (no units); $|$: given.

Given model $P(H) = .01$: prevalence; $P(+ \mid H) = .9$: sensitivity; $P(+ \mid \neg H) = .1$: false-positive rate. Expected counts in 10,000 trials: 100 event cases, 9,900 non-event cases.

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

H event; $+$ positive reading.

Prior probability; posterior probability — [1] DL §3.11 · [9] Wisconsin Math225 Lec.5 · Course example

EXERCISE

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $\mid$: given.

A. .9: the chance of a true event equals sensor sensitivity.

B. .1: an observation cannot change the prior.

C. .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

RECAP

Break

Pause: probability, density, and conditioning

$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

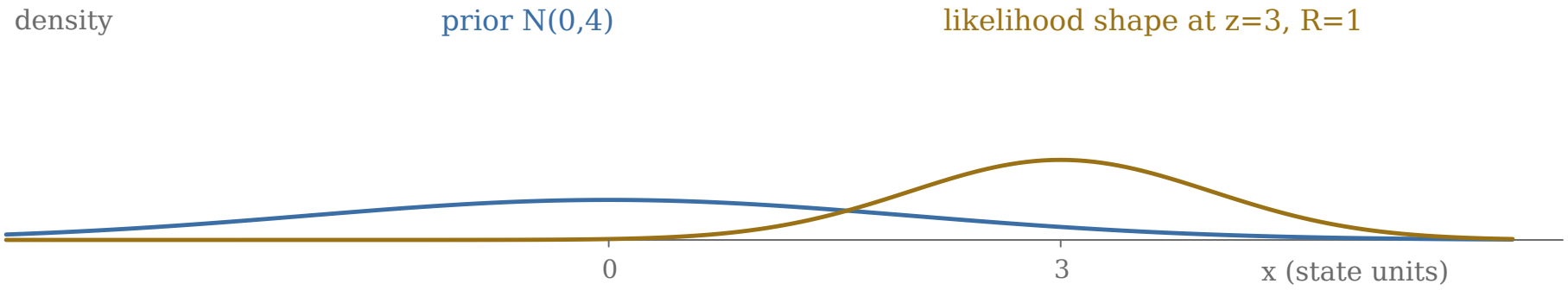
X: random quantity (written x earlier); lowercase x: evaluation value here. A: set of values; $\{X \in A\}$: event; p: density. H: event of interest; $+$: positive reading. Conditioning changes the reference population.

Event probability; density; conditional probability — [1] DL §3.3.2 · [1] DL §3.4–3.5 · Course recap

MODEL

Gaussian conditioning

A posterior density describes uncertainty after a reading.



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

- x, z, ε — unknown random scalar state, scalar reading (fixed after observation), and independent zero-mean Gaussian noise; all in state units.
- μ_0, P_0, R — fixed prior mean (state units), prior variance and noise variance (squared state units); $P_0, R > 0$. $\mathcal{N}$: normal distribution, with mean then variance.
- $p(x), p(z \mid x), p(x \mid z)$ — prior density, likelihood for candidate x with observed z fixed, and posterior density; density units are inverse state units.

Gaussian prior; Gaussian likelihood — [2] UBC Lec.18 p.7 · [8] CMU Bayes notes p.1

DERIVATION

Gaussian conditioning

Precision is inverse variance.

$$J(x) = \frac{(x-\mu_0)^2}{2P_0} + \frac{(z-x)^2}{2R}, \quad P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(\mu_0/P_0 + z/R)$$

- x, z, J unknown scalar state, fixed observed reading, and dimensionless negative log posterior up to an x-independent constant.
- μ_0, P_0, R fixed prior mean, prior variance and independent noise variance; $P_0, R > 0$.
- μ_1, P_1 posterior mean and variance. Means/readings use state units; variances use squared state units.

$$a = P_0^{-1} + R^{-1}, \qquad b = \mu_0/P_0 + z/R$$
$$J(x) = \tfrac{1}{2}(ax^2 - 2bx) + C = \tfrac{a}{2}(x - b/a)^2 + C\prime$$

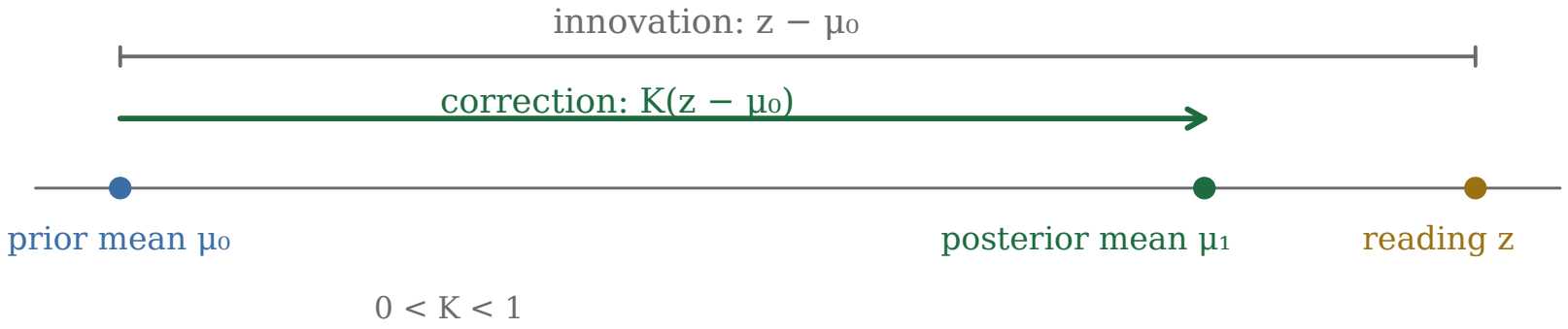
- a, b total precision (inverse squared state units), and precision-weighted location (inverse state units); $C, C\prime$: scalar constants independent of x.

Gaussian posterior; precision — [2] UBC Lec.18 p.7 · [1] DL §3.9.3 · Course derivation

INTERPRETATION

Gaussian conditioning

The gain scales the innovation to update the mean.



$$\mu_1 = \frac{R\mu_0 + P_0 z}{P_0 + R} = \mu_0 + K(z - \mu_0), \quad K = \frac{P_0}{P_0 + R}, \quad P_1 = (1 - K)P_0$$

μ_0, z, μ_1	fixed prior mean, observed scalar reading and posterior mean (state units).
P_0, R, P_1	prior, independent noise and posterior variances (squared state units); $P_0, R > 0$.
$K, z - \mu_0$	dimensionless gain $0

Kalman gain; measurement innovation — [5] Welch–Bishop §1 · [2] UBC Lec.18 p.7 · Course derivation

EXAMPLE

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

x, μ_0, z candidate scalar state on the horizontal axis, fixed prior mean and observed reading (state units).

P_0, R fixed prior variance and independent noise variance (squared state units); $\mathcal{N}$: normal, mean then variance.

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

K gain; μ_1, P_1 posterior mean, variance.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course example

EXERCISE

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$; $z = 3$ with independent Gaussian noise, variance R: $1 \to 4$. Compare posterior mean and variance.

x: scalar state; z: fixed reading; R: independent noise variance. $\mathcal{N}$: normal distribution, mean then variance; states/readings use state units and variances use squared units.

A Mean moves toward 0; posterior variance increases.

B Mean moves toward 3; posterior variance decreases.

C Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

RESULT

Gaussian conditioning

A conditional mean is an expectation given an observation.

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

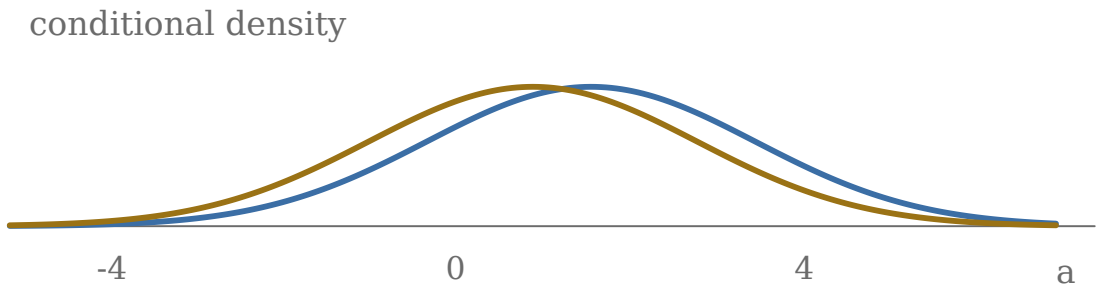
- a, b, p, q — jointly Gaussian random columns of sizes $p \times 1$, $q \times 1$; p, q: coordinate counts. The observed value of b is fixed when conditioning.
- $\mu_a, \mu_b, \mu_{a|b}$ — marginal mean columns (p, q entries), and conditional mean $\mathbb{E}[a \mid b]$ (p entries); means/innovations retain their coordinate units.
- Σ_{ab} — $p \times q$ cross-covariance $\mathbb{E}[(a - \mu_a)(b - \mu_b)^T]$; each entry has the product of the paired coordinate units.
- Σ_{bb} — fixed, invertible $q \times q$ covariance of b; Σ_{bb}^{-1} maps a q-entry innovation to scaled coordinates.

Conditional Gaussian mean — [2] UBC Lec.18 p.7 · [7] Data140 §23.2

RESULT

Gaussian conditioning

Conditional covariance measures spread after an observation.



b = 5: mean 5/3

b = 2: mean 1

both variances 32/9

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

a, b, p, q jointly Gaussian columns with p, q entries; p, q: coordinate counts. In a conditional distribution the observed value of b is fixed.

$\Sigma_{aa}, \Sigma_{a|b}$ $p \times p$ covariance of a before/after observing b; entries have products of a-coordinate units.

$\Sigma_{ab}, \Sigma_{bb}, \Sigma_{ba}$ fixed $p \times q$ cross-covariance, positive-definite $q \times q$ covariance of b, and $q \times p$ transpose $\Sigma_{ba} = \Sigma_{ab}^T$.

Scalar plot a, b are scalars; fixed joint mean $\mu = (1, 2)^T$ and joint covariance $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Horizontal: candidate a; vertical: density (inverse a units).

Conditional Gaussian covariance — [2] UBC Lec.18 p.7 · [2] UBC Lec.18 p.8

EXERCISE

Gaussian conditioning

Does the conditional variance change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

a: unknown scalar; b: observed scalar; μ: fixed 2×1 joint mean; Σ: fixed 2×2 joint covariance. Means/values use coordinate units; covariance entries use paired product units.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

DEFINITION

Estimation

Maximum likelihood estimation (MLE) maximizes data likelihood.

$$b = A\theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

$$L(\theta; b) = p(b \mid \theta),\ \hat{\theta}_{MLE} \in \arg\max_{\theta} L(\theta; b)$$

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, \varepsilon \in \mathbb{R}^N$: data/noise columns; $\theta, \hat{\theta}_{MLE} \in \mathbb{R}^p$: unknown parameter/estimate. $I = I_N$: identity; $\sigma^2 > 0$: known noise variance (squared reading units). $L = p(b \mid \theta)$: data density with observed b fixed; argmax: maximizing values.

Gaussian linear regression; maximum likelihood — [1] DL §5.5

DERIVATION

Estimation

Gaussian products become a squared loss

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}$$

N, p: reading/parameter counts; $A \in \mathbb{R}^{N\times p}$: fixed design; $b \in \mathbb{R}^N$: observed data; $\theta \in \mathbb{R}^p$: varied parameter. $r = b - A\theta \in \mathbb{R}^N$: residual; $\|r\|^2$: sum of squared residuals. Independent errors share known variance $\sigma^2 > 0$ in squared reading units; $p(b \mid \theta)$: data density.

Negative log-likelihood; least squares — [1] DL §5.5.1 · Course derivation

DERIVATION

Estimation

Normal equations set the least-squares gradient to zero.

$$A^T A\hat{\theta} = A^T b, \qquad A^T(b - A\hat{\theta}) = 0$$

N, p: reading/parameter counts; $A \in \mathbb{R}^{N\times p}$: fixed design; $b \in \mathbb{R}^N$: fixed data; $\theta, \hat{\theta} \in \mathbb{R}^p$: candidate/fitted parameters. $r = b - A\hat{\theta} \in \mathbb{R}^N$: fitted residual, in reading units. Independent equal-variance Gaussian errors give the MLE. Full column rank gives uniqueness; $A^T r = 0$ means orthogonality to every design column.

Normal equations; least squares — [1] DL §5.1.4 · Course derivation

EXAMPLE

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$A \in \mathbb{R}^{3\times 2}$: fixed design; $b \in \mathbb{R}^3$: observed readings. $\theta = (\theta_1, \theta_2)^T$: intercept/slope; $\hat{\theta} \in \mathbb{R}^2$: fitted coefficients; $r = b - A\hat{\theta} \in \mathbb{R}^3$: observed minus fitted. Inputs $0, 1, 2$ are dimensionless; both coefficients and residuals use reading units.

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

$\hat{\theta}$ estimate; r observed minus fitted.

Least-squares estimate — [1] DL §5.1.4 · Course example

DEFINITION

Estimation

Maximum a posteriori estimation (MAP) maximizes the posterior density.

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \quad \lambda^2 = \sigma^2/\tau^2$$

N, p: reading/parameter counts; fixed $A \in \mathbb{R}^{N\times p}$ and observed $b \in \mathbb{R}^N$; $\theta, \mu_0 \in \mathbb{R}^p$: candidate parameter/fixed prior mean. Independent Gaussian noise has known variance $\sigma^2 > 0$. Prior: $\mathcal{N}(\mu_0, \tau^2 I_p)$; I_p: identity; $\tau^2 > 0$: variance in each parameter direction. J: scaled negative-log posterior; λ^2: prior weight. Data/parameter variances use their respective squared units.

Maximum a posteriori (MAP); quadratic regularization — [1] DL §5.6.1 · Course derivation

EXAMPLE

Estimation

The prior changes the normal equation

$$(A^TA + \lambda^2 I)\hat{\theta} = A^Tb + \lambda^2\mu_0$$

$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 2}$: fixed design; $b = (1, 2, 2)^T$: observed data. $\theta = (\theta_1, \theta_2)^T$: intercept/slope; $\hat{\theta}$: fitted column; $I = I_2$: identity. Fixed prior mean $\mu_0 = (0, 0)^T$; weight $\lambda^2 = 1$. Inputs are dimensionless; coefficients and residuals use reading units.

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \|b - A\hat{\theta}\|^2 = \tfrac{2}{5}$$

$\hat{\theta}_{MAP} = (\text{intercept}, \text{slope})^T$: posterior mode; $\|b - A\hat{\theta}\|^2$: squared residual norm.

MAP normal equations; ridge regularization — [1] DL §5.6.1 · [1] DL §5.1.4 · Course derivation

EXERCISE

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$. A: fixed 2×2 design; b: observed 2-entry column; $\theta = (\theta_1, \theta_2)^T$: varied parameter; J: scalar objective in the stated numerical coordinates.

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

EXERCISE

Sensor Calibration

Calibration: choose the likelihood

$b = (1, 2, 3)^T$: readings minus a known reference; $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model? $i = 1, 2, 3$: reading index; b: observed 3-entry column; θ: scalar offset; ε_i: random error in reading units.

A $\frac{1}{2} \sum_i (b_i - \theta)^2$

B $\sum_i |b_i - \theta|$

C $\frac{1}{2} \sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE

Sensor Calibration

Calibration: combine data and prior

Readings $(1, 2, 3)$; independent normal noise, variance 1; prior $\theta \sim \mathcal{N}(0, 1)$. Find posterior mean and precision (inverse variance). $b = (1, 2, 3)^T$: fixed data column; θ: scalar additive offset, in reading units; normal notation gives mean then variance.

A. Center stays at 2; only uncertainty changes.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE

Sensor Calibration

Calibration: confidence coverage

Three iid $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; $\bar{B}$: mean. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. Interpret 95% confidence. $B = (B_1, B_2, B_3)^T$: random data; θ: fixed scalar offset; $\bar{B}$: random average; $C(B)$: interval rule. Offset and endpoints use reading units.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE

Sensor Calibration

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability? b: fixed observed 3-entry column; θ: scalar offset. Normal notation lists mean then variance; 1.5 is in reading units and $.25$ in squared units.

A. 95% posterior mass for θ, with b fixed.

B. 95% coverage for every fixed true offset.

C. 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE

Sensor Calibration

Calibration: why do the intervals differ?

Prior $\theta \sim \mathcal{N}(0, 1)$. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$. Why different? $b = (1, 2, 3)^T$: fixed data; θ: scalar offset; normal noise variance 1. $C(b), I(b)$: the two displayed interval constructions, in reading units.

A. Narrower credible interval means better coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE

Sensor Calibration

Calibration: new data, new prior

Readings $b = (2, 3)^T$ estimate offset θ; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which is valid? b: fixed observed 2-entry column; θ: scalar additive offset in reading units; normal notation lists mean then variance.

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE

Sensor Calibration

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE

Synthesis

Which modeling bridge is valid?

Zero-mean iid Gaussian noise; isotropic Gaussian prior. Which chain gives the negative-log posterior?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; $-\log$ turns independent factors into sums.

C. A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

References

Sources and optional extensions

[1] Deep Learning, chs.2–5 (2016).

[2] UBC CS340: Gaussian Distributions (2011).

[3] NIST/SEMATECH Statistical Handbook.

[4] MIT18.05: Probability Intervals (2022).

[5] Welch & Bishop: Kalman Filter (2004).

Definitions: [6] Stanford EE278 · [7] Data140 · [8] CMU Bayes notes · [9] Wisconsin Math225 · [10] MC Probability